From: Chloe Johnson
Sent: Thursday, July 20, 2023 3:16 PM
To: Blong, Kamon Van; Ryan Morningstar
Cc: Radulovich, Andrew David; Hooda, Pavit Mohan
Subject: Re: New submission from Contact Us

Some people who received this message don't often get email from chloe@elementum3d.com. Learn why this is important

---- **External Email**: Use caution with attachments, links, or sharing data ----

Hi Kamon,

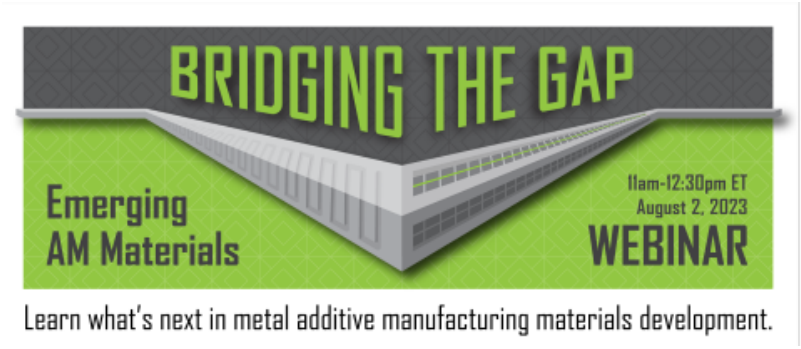
Just wanted to go ahead and send this over before I forgot. See the table below for the HIP & T6 A6061-RAM2 data. I'll see if we have any thin wall measurements for you for the 40 um to compare to the 30 um I sent over last time.

A6061-RAM2 (HIP & T6)			
Temperature (° C)	Specific Heat (J/g·K)	Diffusivity(mm2/s)	Thermal Conductivity (W/m·K)
25	1.053	66.23	187.5
40	1.135	66.53	204.3
60	1.146	66.78	203.5
80	1.133	67.29	200
100	1.169	67.41	207.7
120	1.174	67.56	205.6
140	1.219	67.73	218
160	1.22	67.82	216.9
180	1.232	67.84	218.4

Let me know if you have any other questions.

Thanks,

Chloe Johnson, PhD
Materials Engineer
400 Young Ct., Unit 1
Erie, CO 80516
+1 720-263-9513



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From: Blong, Kamon Van <kblong@purdue.edu>
Sent: Thursday, July 20, 2023 8:18 AM
To: Ryan Morningstar <ryanm@elementum3d.com>; Chloe Johnson <Chloe@elementum3d.com>
Cc: Radulovich, Andrew David <aradulov@purdue.edu>; Hooda, Pavit Mohan <phooda@purdue.edu>
Subject: Re: New submission from Contact Us

This email is external to Elementum 3D
Ryan,

Yes, 2:30est also works for us. We look forward to talking with you then!

Best,
—

Kamon Blong
Purdue University
College of Engineering | AAE '25
765-469-2579

From: Ryan Morningstar <ryanm@elementum3d.com>
Sent: Wednesday, July 19, 2023 2:34 PM
To: Blong, Kamon Van <kblong@purdue.edu>; Chloe Johnson <Chloe@elementum3d.com>
Cc: Radulovich, Andrew David <aradulov@purdue.edu>; Hooda, Pavit Mohan <phooda@purdue.edu>
Subject: New submission from Contact Us
When: Thursday, July 20, 2023 2:30 PM-3:00 PM.
Where: Microsoft Teams Meeting

Some people who received this message don't often get email from ryanm@elementum3d.com. Learn why this is important

---- **External Email**: Use caution with attachments, links, or sharing data ----

Hello Kamon,

Does this time work for you? I won't be able to make the later time. We could also do 5pm ET

Regards,
Ryan

Hello Kamon,

Does this time work for you to meet?

Regards,
Ryan

Microsoft Teams meeting

Join on your computer, mobile app or room device
Click here to join the meeting

Or call in (audio only)
+1 720-912-5154,,448788540# United States, Denver

Phone Conference ID: 448 788 540#
Find a local number | Reset PIN

If you are joining from outside of the Azure Government/GCC-H environment please open the meeting link in a web browser and join as guest.

Learn More | Meeting options

From: Blong, Kamon Van <kblong@purdue.edu>
Sent: Monday, July 17, 2023 9:28 AM
To: Ryan Morningstar <ryanm@elementum3d.com>
Cc: Radulovich, Andrew David <aradulov@purdue.edu>; Hooda, Pavit Mohan <phooda@purdue.edu>
Subject: Re: New submission from Contact Us

This email is external to Elementum 3D
Good afternoon Ryan,

Thanks for getting back to us! Before talking more on pricing, we would ideally like to further discuss our requirements, Elementum capabilities, and our team to make sure everything's a good fit. Is there any way you or another member of your team would be available sometime in the next week or so to meet?

On a similar note, in order to finalize a design with A6061-RAM2 and check it meets our requirements, we would need more information about the alloy and your printing capabilities beyond what I could find on the alloy data sheet. Specifically, we were looking for specifications on printing tolerances / minimum feature sizes as well as thermal conductivity at temperature or strength past 250C if you have that information.

All of the design suggestions you made look like they would work just fine on our side and should be quick changes. Would you have any information on overhang features / design for manufacturability for your printers (mostly on what radius to make rounded upper corners before they turn into an overhang versus a corner)?

Thanks for the help!

Best,

—

Kamon Blong
Purdue University
College of Engineering | AAE '25
765-469-2579

From: Ryan Morningstar <ryanm@elementum3d.com>
Sent: Monday, July 10, 2023 7:49 PM
To: Blong, Kamon Van <kblong@purdue.edu>
Cc: Radulovich, Andrew David <aradulov@purdue.edu>; Hooda, Pavit Mohan <phooda@purdue.edu>
Subject: RE: New submission from Contact Us

---- **External Email**: Use caution with attachments, links, or sharing data ----

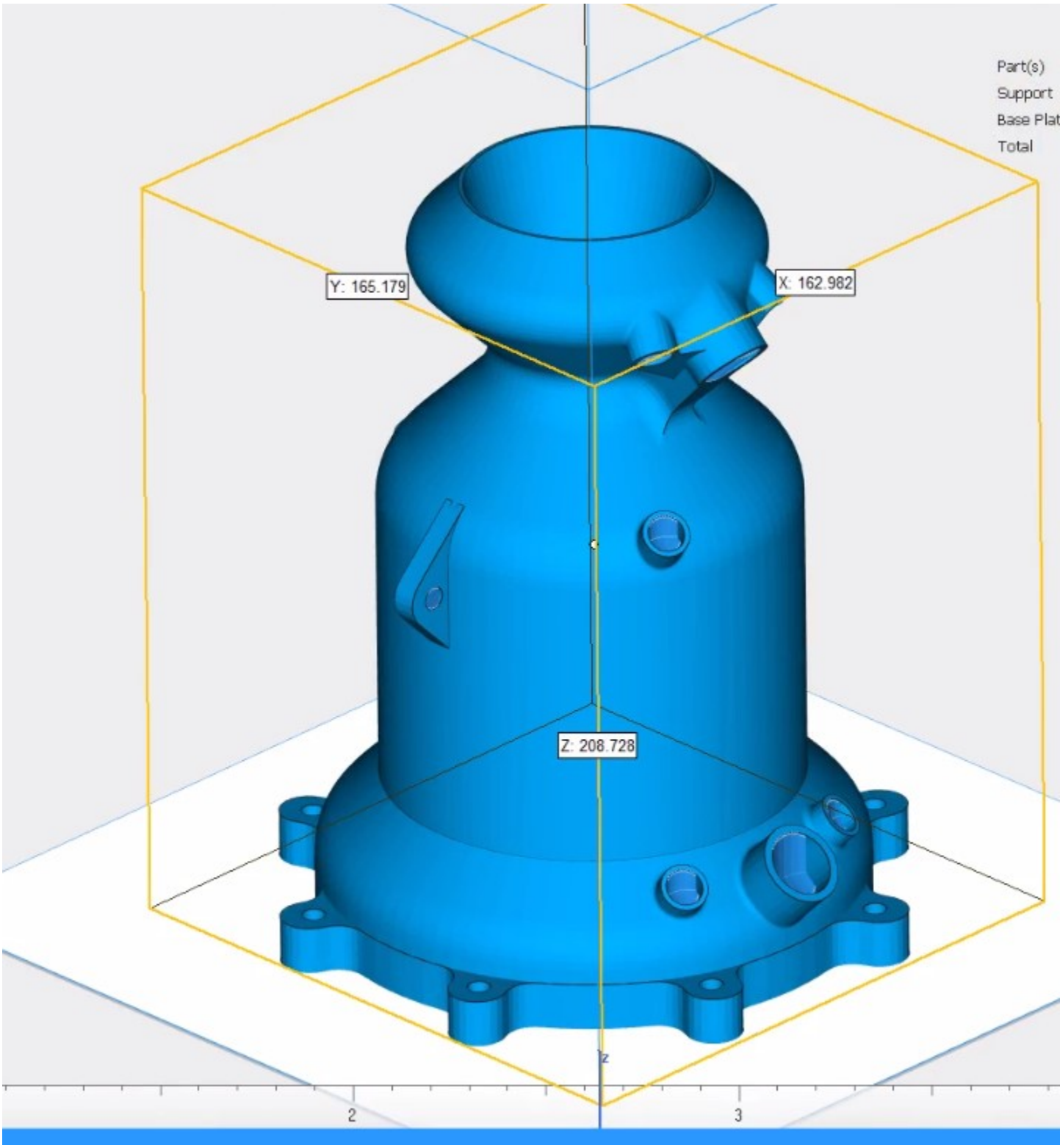
Hello Kamon,

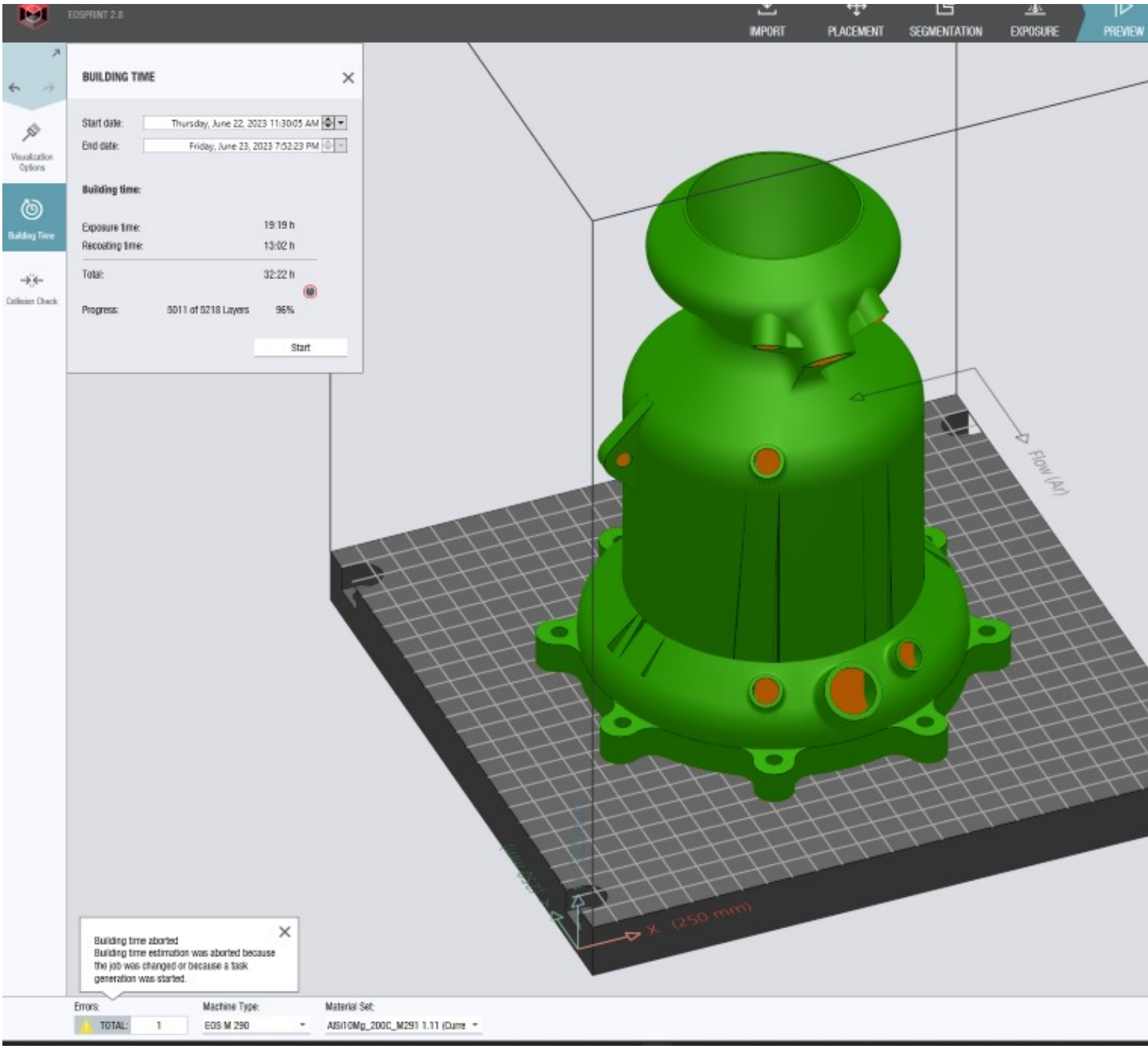
Thanks for checking back in, the quote is ready but had slipped past me on sending it out.

If we were to charge normal pricing it would be $5754 to print the part. How much would you be able to cover? I will need to propose the remainder to our management to get approval.

Also, there are a few printability design suggestions we have on the part, would these be able to be incorporated?

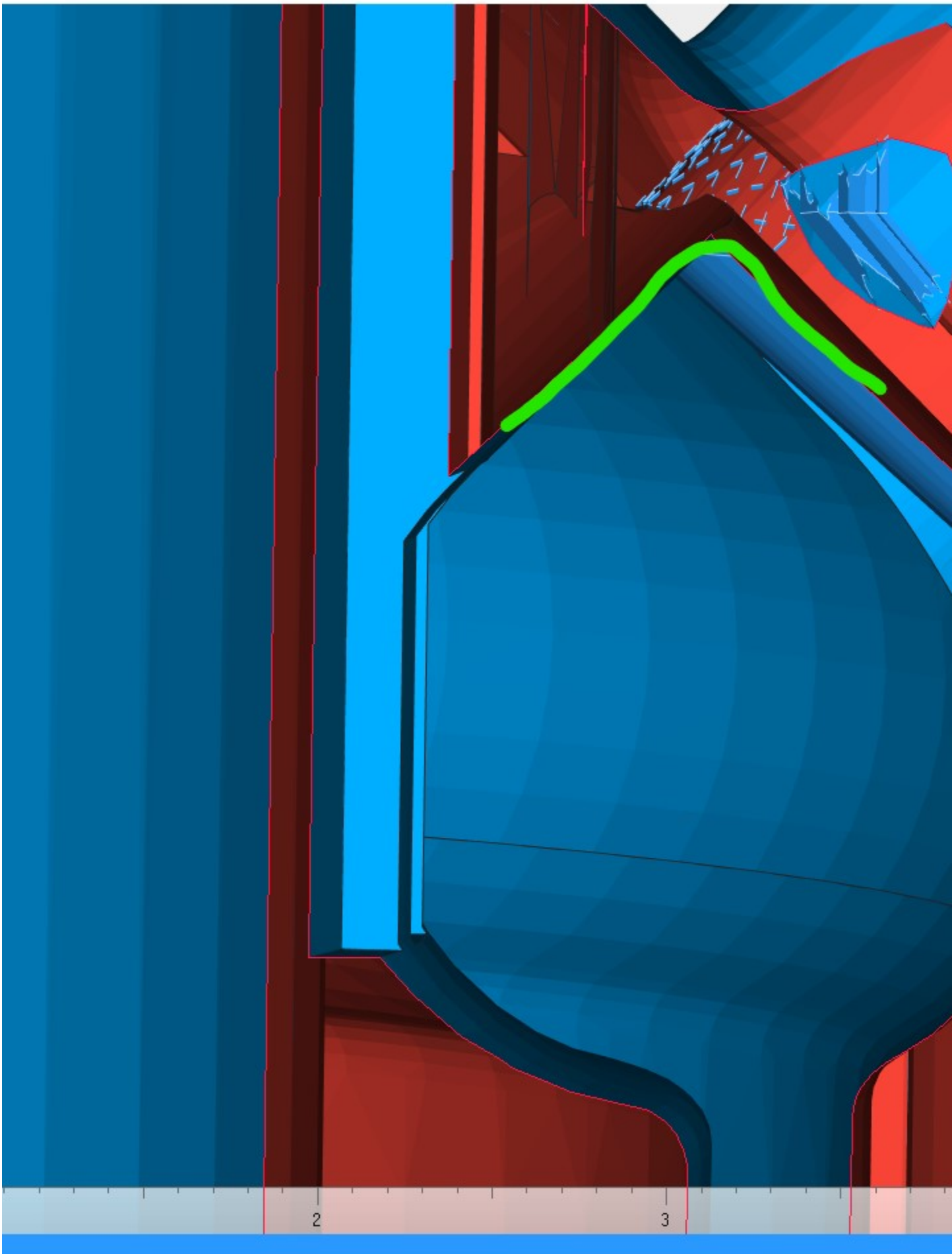
- A6061-RAM2, 40um settings used
- Bandsaw part removal
- No support removal


Part(s)
Support
Base Plat
Total
Y: 165.179
X: 162.982
Z: 208.728
2
3



Might need additional Design For Additive Manufacturing alterations to the CAD file. See below for suggestions:**

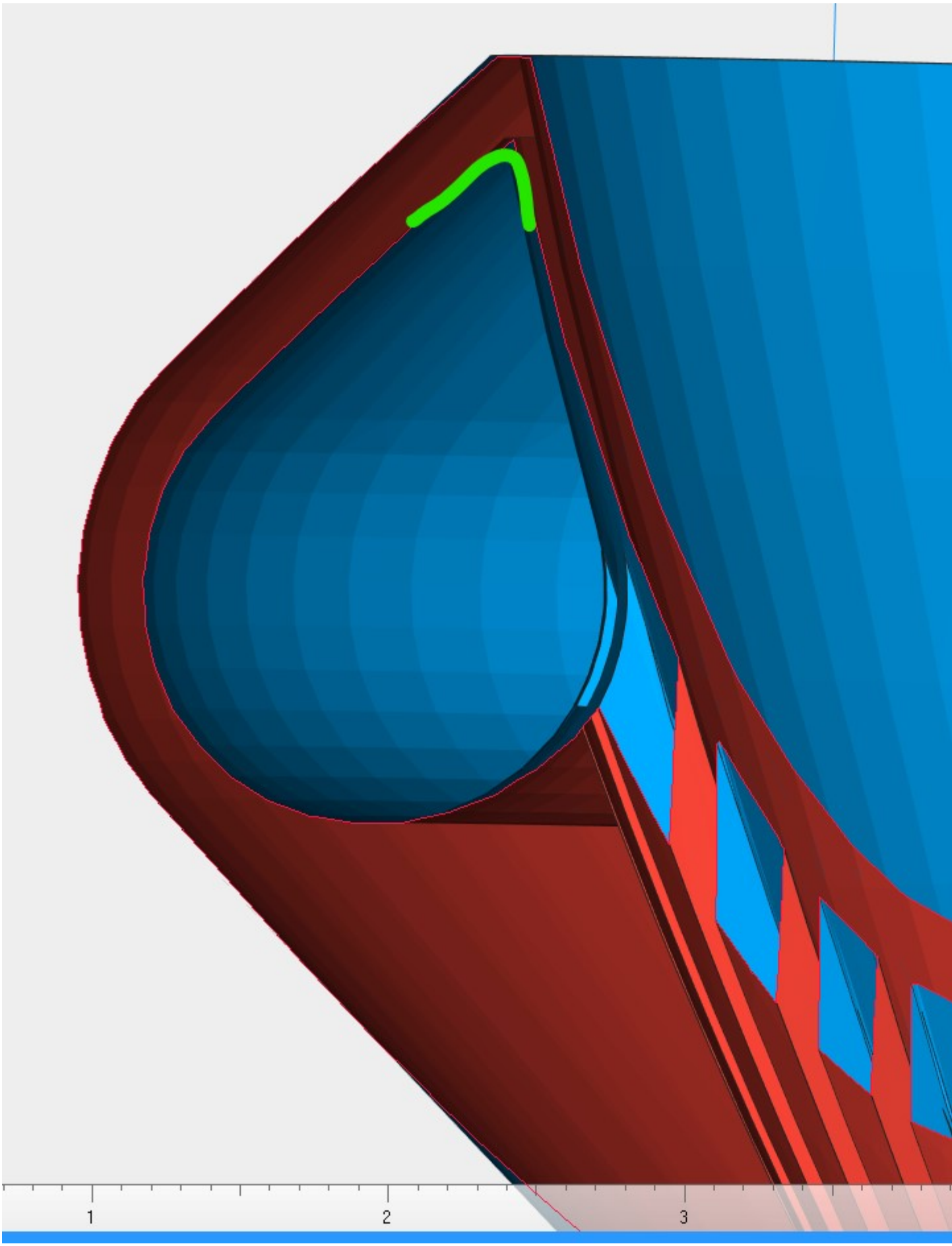
- Rounding of interior chamber roof, instead of a sharp angle.


2
3

- Rounding (arches) of all interior vertical channeling. This is the main portion that we would like to see fixed.


2
3
4

- Rounding of sharp intersection on upper interior chamber.


1
2
3

Regards,

Ryan Morningstar
Account Manager - Elementum 3D
(720) 810-1611

From: Blong, Kamon Van <kblong@purdue.edu>
Sent: Monday, July 10, 2023 8:05 AM
To: Ryan Morningstar <ryanm@elementum3d.com>
Cc: Radulovich, Andrew David <aradulov@purdue.edu>; Hooda, Pavit Mohan <phooda@purdue.edu>
Subject: Re: New submission from Contact Us

This email is external to Elementum 3D
Good morning Ryan,

I just wanted to follow up to see if you had a chance to read this email / send our CAD file through for a quote.

Best,
—

Kamon Blong
Purdue University
College of Engineering | AAE '25
765-469-2579

From: Blong, Kamon Van <kblong@purdue.edu>
Sent: Tuesday, June 20, 2023 3:31 PM
To: Ryan Morningstar <ryanm@elementum3d.com>
Subject: Re: New submission from Contact Us

Ryan,

Nice to meet you and thank you for your prompt response! Apologies for the delay on my end; we've been reviewing your alloys and looking at which one would be best for our application.

For our engine, the main properties we are looking for are high thermal conductivity and strength at temperature, as well as the ability to print small features. From a preliminary review, it looks like A6061-RAM2 would be our preferred alloy, but I was unable to find any specifications for printing tolerances / minimum feature sizes if you'd be able to share any of that information. Additionally, if you would have any charts you could send for thermal conductivity at temperature (such as the one in A1000-AM sheet), that would be greatly appreciated.

I've attached a copy of our sponsorship packet and CAD below. Our current model was designed for AlSi10Mg on an EOS machine, so we will need to make some small changes, but the overall form factor will remain the same.

Very cool to hear about the NASA engine made from A6061 though - I've worked with some of your GRCop hardware in the past so I know you definitely manufacture a high quality product!

Please let me know if you have any questions or need any more information - we look forward to hearing back from you!

Best,

—

Kamon Blong
Purdue University
College of Engineering | AAE '25
765-469-2579

From: Ryan Morningstar <ryanm@elementum3d.com>
Sent: Friday, June 16, 2023 10:18 AM
To: Blong, Kamon Van <kblong@purdue.edu>
Subject: RE: New submission from Contact Us

> You don't often get email from ryanm@elementum3d.com. Learn why this is important

---- **External Email**: Use caution with attachments, links, or sharing data ----

Hello Kamon,

Nice to emeet you, I'm an account manager here with a metallurgy background.

That's great to hear you're looking at Aluminium engines! We just had NASA Hot Fire a 60,000 lbf engine made of our A6061-RAM2.

We're interested to partner with you. Please send over the information when you get a chance!

We'll review and quote it out, then talk over cost sharing. Do you know what alloy you'd like or would you prefer we suggest one?

Regards,

Ryan Morningstar
Account Manager - Elementum 3D

(720) 810-1611

From: Kamon Blong kblong@purdue.edu <noreply@elementum3d.com>
Sent: Wednesday, June 14, 2023 9:37 PM
To: Information <info@elementum3d.com>
Subject: New submission from Contact Us

This email is external to Elementum 3D

Name
Kamon Blong
Email
kblong@purdue.edu
Message
Hello, My name is Kamon Blong, and I am contacting you as a lead of Purdue Space Program: Active Controls, a student organization at Purdue University. Our team is working on a self landing liquid rocket and pushing the bounds of what has been accomplished by collegiate teams worldwide. Our first step in this project is the development of a water-cooled rocket engine which will be 3d printed out of a heat-treated aluminum alloy, and we would like to approach Elementum 3D as a potential sponsor for this project. Due to our limited resources as a student team, we would likely not have the ability to front a full quote for this part and as such would require a sponsorship to enable completion of our project. However, we believe this would be an excellent showcase of your printing capabilities and would look forward to fostering a relationship in the future. I can provide more detail on our design requirements, current CAD of our design, and a team sponsorship packet via my email kblong@purdue.edu. We would love to discuss our development plans and how we could work together! Thanks, Kamon Blong